

Ch 12 HW #2

Due: 11:59pm on Monday, August 31, 2026

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Problem 12.70

A particle moves along the curve $y = e^{2x}$ such that its velocity has a constant magnitude of $v = 5 \text{ ft/s}$.

Part A

Determine the x component of velocity when the particle is at $y = 4 \text{ ft}$.

Express your answer in feet per second to three significant figures.

ANSWER:

$v_x =$ ft/s

Part B

Determine the y component of velocity when the particle is at $y = 4 \text{ ft}$.

Express your answer in feet per second to three significant figures.

ANSWER:

$v_y =$ ft/s

Problem 12.72

A particle travels along the circular path $x^2 + y^2 = r^2$.

Part A

If the y component of the particle's velocity is $v_y = 2r \cos(2t)$, determine the x and y components of its acceleration at any instant.

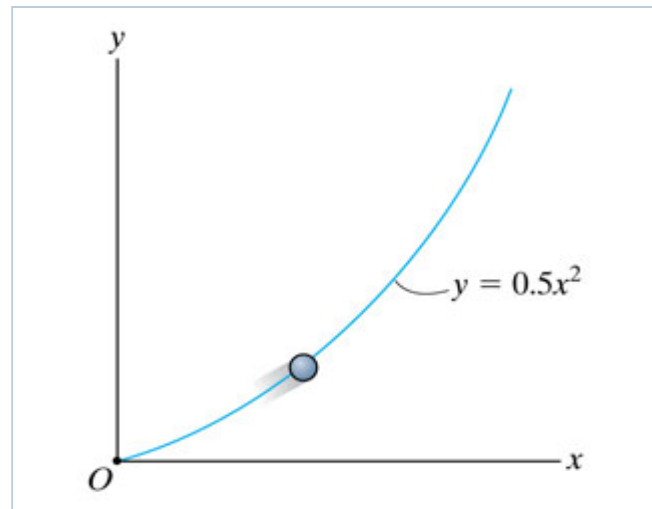
Express your answers in terms of the variables r and t . Enter the x and y components of the acceleration separated by a comma.

ANSWER:

$a_x, a_y =$

Problem 12.79

The particle travels along the path defined by the parabola $y = 0.5x^2$, where x and y are in ft. When $t = 0$, $x = 0$, $y = 0$.



Part A

If the component of velocity along the x axis is $v_x = (5t)$ ft/s, where t is in seconds, determine the particle's distance from the origin O when $t = 2$ s.

Express your answer using three significant figures and include the appropriate units.

ANSWER:

$d =$

Part B

Determine the magnitude of the particle's acceleration when $t = 2$ s.

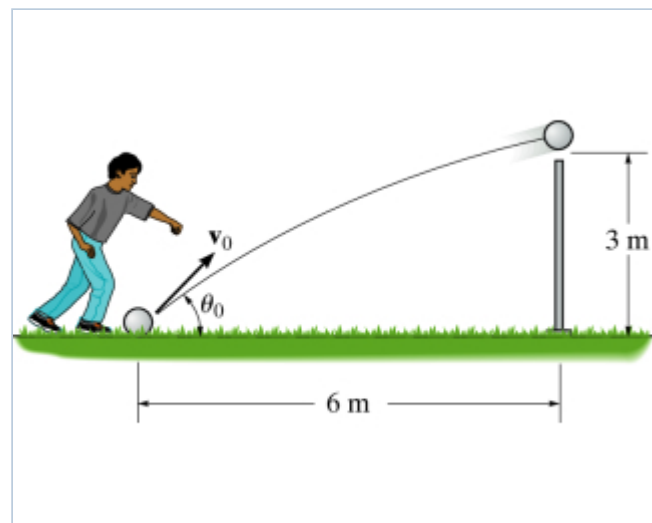
Express your answer using three significant figures and include the appropriate units.

ANSWER:

$a =$

Problem 12.86

Consider .



Part A

Determine the minimum initial velocity v_0 at which the ball must be kicked in order for it to just cross over the 3-m high fence.

Express your answer using three significant figures and include the appropriate units.

ANSWER:

$v_0 =$

Part B

Determine the corresponding angle θ_0 at which the ball must be kicked.

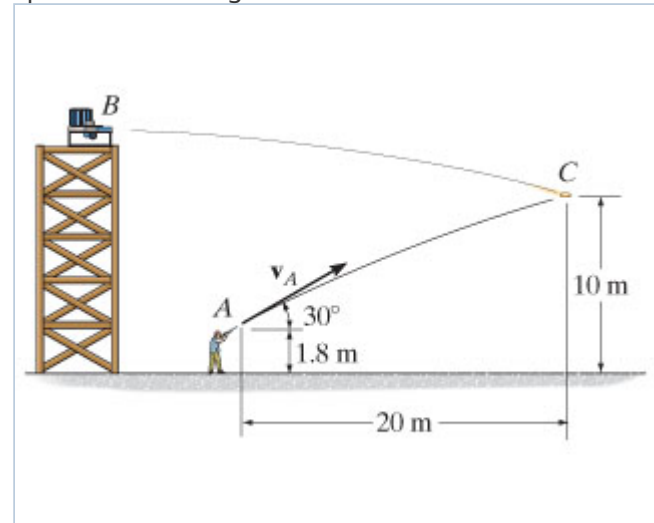
Express your answer in degrees using three significant figures.

ANSWER:

$\theta_0 =$

Problem 12.99

A projectile is fired from the platform at B . The shooter fires his gun from point A at an angle of 30° .



Part A

Determine the muzzle speed of the bullet if it hits the projectile at C .

Express your answer to three significant figures and include the appropriate units.

ANSWER:

$v =$

Score Summary:

Your score on this assignment is 0.0%.

You received 0 out of a possible total of 5 points.